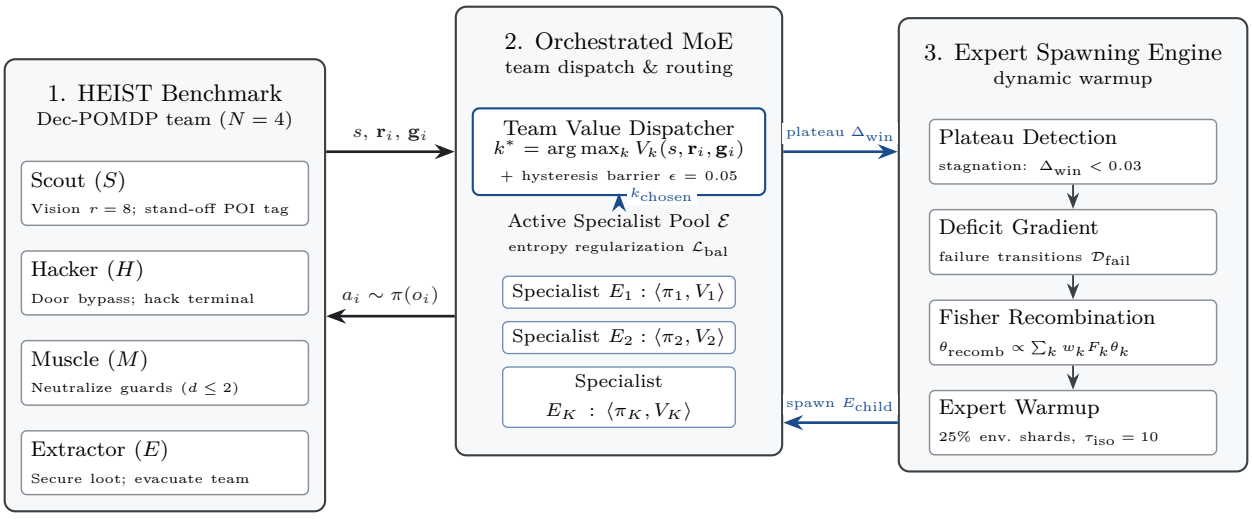
1. HEIST Benchmark
Dec-POMDP team ($N = 4$)
Scout ($S$)
Vision $r = 8$; stand-off POI tag
Hacker ($H$)
Door bypass; hack terminal
Muscle ($M$)
Neutralize guards ($d \leq 2$)
Extractor ($E$)
Secure loot; evacuate team
$s,\ \mathbf{r}_i,\ \mathbf{g}_i$
$a_i \sim \pi(o_i)$
2. Orchestrated MoE
team dispatch & routing
Team Value Dispatcher
$k^* = \arg\max_k V_k(s, \mathbf{r}_i, \mathbf{g}_i)$
+ hysteresis barrier $\epsilon = 0.05$
$k_{\text{chosen}}$
Active Specialist Pool $\mathcal{E}$
entropy regularization $\mathcal{L}_{\text{bal}}$
Specialist $E_1 : \langle \pi_1, V_1 \rangle$
Specialist $E_2 : \langle \pi_2, V_2 \rangle$
Specialist
$E_K : \langle \pi_K, V_K \rangle$
plateau $\Delta_{\text{win}}$
spawn $E_{\text{child}}$
3. Expert Spawning Engine
dynamic warmup
Plateau Detection
stagnation: $\Delta_{\text{win}} < 0.03$
Deficit Gradient
failure transitions $\mathcal{D}_{\text{fail}}$
Fisher Recombination
$\theta_{\text{recomb}} \propto \sum_k w_k F_k \theta_k$
Expert Warmup
25% env. shards, $\tau_{\text{iso}} = 10$